

RECSYS CHALLENGE 2026 · MUSIC-CRS

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

Nidhin Pattaniyil · Semih Yagli · Tanwir Zaman

Independent Researchers, Hoboken, New Jersey, USA · team npatta01

47,071

tracks in the catalog

11

ways to search

1

learned ranker





The short version

SUMMARY

TALK ONLY

WHAT WE BUILT

An LLM reads each turn into a **typed state**; eleven retrieval branches search one index; a **LightGBM** ranker picks 20; an LLM writes the reply.

HOW IT SCORED

0.3811

Blind-B · 29th of 40
nDCG@20 fell from 0.438 on Blind-A to 0.254.

WHAT WE FOUND

31 of 43

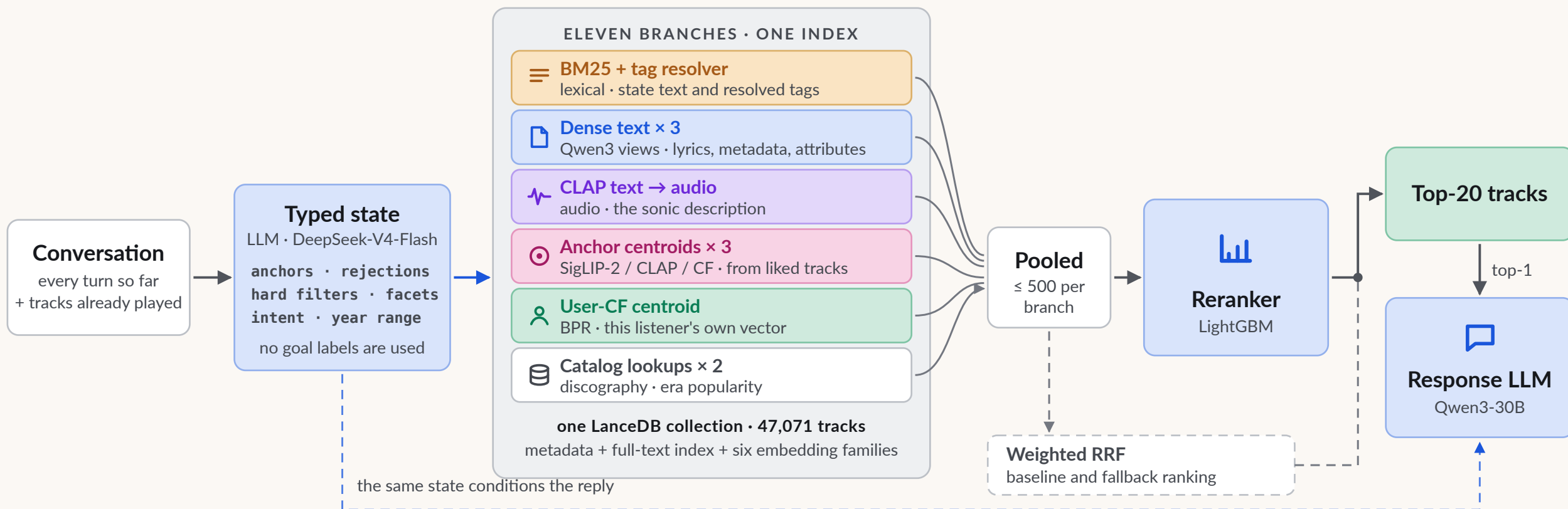
diagnosed failures were ranking

Extraction was **3**. The state was richer than what the ranker used.

The system understood the conversation more often than it **acted on it**.



One state, eleven branches, one learned ranker



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.



Request understanding

TURN 1

🎵 “Calling Cards” — *not quite*

TURN 2

🎵 “Man” — *not it either*

TURN 3 · NOW

Much more subdued... a sense of place in the lyrics... almost a cappella.

LLM



STAY WITH

Neko Case

TURNED DOWN

“Calling Cards”, “Man”

MOOD

subdued

LYRICS

a sense of place

SOUND

stark, almost a cappella

The LLM reads **every turn so far** and writes structured fields; every later step reads those.



Retrieval — eleven branches

APPROACH

CORE

×1



BM25 + tag resolver

keywords and resolved tags
from the state

×3



Dense text

Qwen3 embeddings of lyrics,
metadata, attributes

×1



CLAP text → audio

the sonic description, matched
to audio

×3



Anchor centroids

near liked tracks: cover art,
audio, listening

×1



User-CF centroid

this listener's own BPR vector

×2



Catalog lookups

the artist's discography; era
popularity

Each branch returns its **top 500**; the ranker sees their union.



What the ranker leans on

FEATURE FAMILY	#	GAIN
Bi-encoder cosine	1	59.0%
Per-branch rank and score	66	11.5%
Session and artist affinity	18	10.7%
Popularity · state · tags	31	9.2%
Other similarity cosines	17	8.9%
Constraints and rejections	13	0.7%

SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.



How it scored

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.

⚠️ Extracted, but the ranker did not act on it

WENT WRONG CORE



*"I gotta break away from **Deltron** and **Del** completely"*

Dead Prez — Hip Hop

state held both artists as avoid

NOT ACTED ON



*"branching out to **other** singer-songwriter artists"*

Ryan Adams — When The Stars Go Blue

state held Ryan Adams as avoid

NOT ACTED ON



*"still focusing on **Juanes**... haven't found the depth I'm after"*

Juanes — Me Enamorado De Ti

state held Juanes as rejected

NOT ACTED ON

We wanted the ranker to use what the state recorded; it served that artist anyway. We think the labels taught it to: they often repeat the just-played artist, so rejections carried only **0.7%** of its decisions.



Where the failures were — all 80 Blind-B turns

DIAGNOSED GAP	TURNS	MEANING
Ranking	31	a better, non-violating candidate sat below our top-20
Constraint not applied	8	an extracted filter never reached the pool
State extraction	3	the extractor missed something the user said
Entity resolution	1	a name never resolved to a catalog ID

TOP-1 FLAGGED

17 / 80

the track we led with violated something the state had recorded

HOW THIS WAS COUNTED

43 diagnoses across 32 distinct turns. A **rule-based check** of each turn’s system trace — not an LLM, and not a relevance label.

Extraction is **3 of 43** diagnoses. The failure is overwhelmingly downstream of understanding the conversation.



Asked for, served

WENT WRONG

TALK ONLY

1

*“the Kamelot track from **Silverthorn**”*

Kamelot — Fallen Star

from *Haven*. We extracted “Silverthorn” but never filtered on it.

CONSTRAINT

2

*“exactly **126.70 bpm**, in **G minor**”*

no tempo or key in the catalog

and we added no data from outside it

NO DATA

3

*“**‘Watercolors’** by **Pat Metheny**”*

Pat Metheny — Alfie

reply: “Got it — here’s ‘Alfie’...” — “Watercolors” is not in the catalog

RESPONSE



What the training labels look like

WENT WRONG

CORE



*“I keep asking for **Arcade Fire** — that's the only song I want”*

label: **Arctic Monkeys — This House is a Circus**

DOES NOT



“similar chill vibe, but from a different artist”

label: **Bonobo — Jets**

MOVES



*“what **other artists** fit this darkwave style?”*

label: **Tropic of Cancer — The Dull Age**

MOVES



*“what **other** orchestral tracks would you recommend?”*

label: **Nick Phoenix — Kogan**

MOVES

Three of four are annotated **MOVES TOWARD GOAL** — the dataset records them as having helped. Train on this and the model learns to anchor. **Ours does.**

Why: the answer is the simulated recommender's pick from a 16–32-track pool, made without seeing the listener's goal.



Everything is released

THANK YOU

CORE



Paper, poster, slides, code, models, data

npatta01.github.io/music-crs-2026